Critical Care Medicine
April 2026 • Volume 54 • Number 4 • Pages 950–959

REVIEW ARTICLE

Approaches to Converting Spo_2/Fio_2 Ratio to Pao_2/Fio_2 Ratio for Assessment of Respiratory Failure in Critically Ill Patients: A Systematic Review

Dipayan Chaudhuri, MD[1,2]
Julieta Lazarte, MD, PhD[1]
Kamya Shah[3]
Tyler Pitre, MD[4]
Pirkka T. Pekkarinen, MD, PhD[5]
Cornelius Sendagire, MD[6,7]
Greg S. Martin, MD, MSc[8,9]
Christian Jung, MD, PhD[10,11]
John G. Laffey, MD, DSc[12,13]
Bram Rochwerg, MD[1]
on behalf of the Sequential Organ Failure Assessment (SOFA)-2 study group

OBJECTIVE: The Pao_2/Fio_2 (PF) ratio is widely used as an assessment of respiratory failure in guiding ventilation strategies and prognostication in critically ill patients. However, given that it mandates invasive arterial access, the Spo_2/Fio_2 (SF) ratio has been suggested as a noninvasive and readily accessible alternative. What are the best ways to convert SF and PF ratios in critically ill patients, in terms of their diagnostic/prognostic accuracy and clinical utility?

DATA SOURCES: We comprehensively searched databases (MEDLINE, Embase, Web of Science, Cochrane library) to identify relevant studies.

STUDY SELECTION: Any observational studies that compared the SF to PF ratio in critically ill patients. We assessed individual study risk of bias (ROB) using the revised QUADAS II tool.

DATA EXTRACTION: We included 45 observational studies, ranging from 61 to 141,000 measurements.

DATA SYNTHESIS: SF to PF imputation was less accurate when the Spo_2 was equal to or greater than 97%. Otherwise, all studies were able to establish strong correlational relationships between SF and PF ratios, but there was no clear best equation. Based on ease of use, size, generalizability and methodology, we were able to prioritize four equations (one linear, one logarithmic linear, and two nonlinear). All four equations showed strong correlation between SF and PF ratios, with the linear equation being easiest to apply. The SF ratio also correlated well with clinical outcomes when compared with the PF ratio, both as an individual value and as part of a comprehensive score, with more discriminating performance in some cases.

CONCLUSIONS: SF and PF ratios demonstrate good correlation, and may have similar prognostic value. Although there is no clear optimal method to convert SF to PF ratios, linear equations show acceptable correlation and are most easily applied at the bedside.

KEYWORDS: acute respiratory distress syndrome; Pao_2/Fio_2 ratio; Spo_2/Fio_2 ratio; Sequential Organ Failure Assessment; systematic review

Assessment of respiratory failure stands as a cornerstone in the management of critically ill patients. Among the tools available, the Pao_2/Fio_2 (PF) ratio has long been regarded as a fundamental parameter, aiding clinicians in evaluating the severity of respiratory failure and guiding therapy (1). However, given the requirements for invasive arterial access, multiple blood draws and test result lag associated with the PF ratio, recent attention has shifted toward the Spo_2/Fio_2 (SF) ratio as a potential alternative to the PF ratio (2).

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.

DOI: 10.1097/CCM.0000000000007018

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.

KEY POINTS

Question: What are the best ways to convert Spo_2/Fio_2 (SF) and Pao_2/Fio_2 (PF) ratios in critically ill patients, in terms of their diagnostic/prognostic accuracy and clinical utility?

Findings: This systematic review revealed a strong correlational relationship between SF and PF ratios, but there was no clear best equation. Based on ease of use, size, generalizability, and methodology, we were able to prioritize four equations with the linear equation being easiest to apply.

Meaning: SF ratios may represent a justifiable alternative to PF ratios and while there is no clear optimal method to convert SF to PF ratios, linear equations show acceptable correlation and are most easily applied at the bedside.

The PF ratio, calculated by dividing the Pao_2 by the Fio_2, has been widely adopted in clinical practice and research settings as a marker of oxygen exchange efficiency. Its simplicity and previous validation studies have led to its incorporation into acute respiratory distress syndrome (ARDS) and Sequential Organ Failure Assessment (SOFA) score severity thresholds and have contributed to its use in guiding ventilator strategies and prognostication (3).

In contrast, the SF ratio, derived from pulse oximetry (Spo_2) and Fio_2, offers a noninvasive and readily accessible alternative for assessing oxygenation status. By using pulse oximetry, the SF ratio eliminates the need for arterial blood gas sampling, reducing patient discomfort, procedural complications and cost associated with arterial cannulation (2). Further, the SF ratio can provide a continuous output, independent from physician or nurse actions, which can be beneficial in data-driven environments like the ICU. These factors make it an attractive tool for rapid assessment in various clinical settings, including the ward and resource-limited environments where access to arterial sampling may be limited. However, the SF ratio indirectly estimates arterial oxygenation based on pulse oximetry readings, and this may introduce variability and inaccuracies, particularly in patients with altered hemoglobin levels, different metabolic states, altered peripheral perfusion, darker skin tone (4) or in the presence of carboxyhemoglobin. In addition, the accuracy and precision of pulse oximetry can be lower than that of blood gas analysis and can vary based on the type of pulse oximeter (5).

Despite the potential benefits of the SF ratio, it remains unclear whether it can be used interchangeably with the PF ratio, and if so, what is the best methodology. Thus, this review was conducted to assess the feasibility and optimal methodology for incorporating the SF ratio as an alternative to the PF ratio, as part of the SOFA 2.0 project.

STUDY DESIGN AND METHODS

We registered the protocol on PROSPERO Number CRD42024562498 and used the Preferred Reporting Items for Systematic reviews and Meta-Analyses checklist to report findings (**Supplementary Table 1**, https://links.lww.com/CCM/H873).

Data Sources and Searches

We performed a comprehensive search of MEDLINE, Embase, the Cochrane library and Web of Science from inception to October 18, 2023 with the help of an experienced medical librarian. The search was updated on November 5, 2024. A complete copy of the search strategy can be found in **Supplementary Materials** (https://links.lww.com/CCM/H873). We screened reference lists of relevant reviews and contacted experts to ensure that we were not missing any key articles.

Study Selection

Two reviewers (D.C., T.P.) screened citations independently and in duplicate in two stages; first they screened the title and abstract and then, they screened the full text of potentially relevant abstracts. Reasons for study exclusion were captured at the full text review stage. A third author (B.R.) adjudicated disagreements in full text screening, where required.

Eligibility Criteria

We included any observational study involving critically ill patients and comparing SF ratio to PF ratio, either directly by establishing a mathematical correlation or indirectly by comparing their diagnostic accuracy

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.

at predicting mortality, ICU length of stay, mechanical ventilation or use of noninvasive respiratory support. We excluded case series, case reports, review articles and editorials.

Data Abstraction and Quality Assessment

Two independent reviewers, working in pairs, from a group of four investigators (J.L., K.S., D.C., T.P.) abstracted data independently and in duplicate using a standardized abstraction form, that was reviewed and pilot tested by the study's primary author (D.C.). A third reviewer (D.C. or B.R.) adjudicated disagreements, where necessary. We abstracted data reporting on study characteristics, baseline patient demographics and examining the relationship between SF and PF ratios. Specifically, we collected data on correlation between SF and PF using the Spearman rho (R^2) and data on discrimination accuracy into PF groups (< 100, 100–200, 200–300) using sensitivity, specificity and area under the curve (AUC). When examining prognostic utility of the SF ratio, when compared with the PF ratio, we looked again at the R^2 for correlation, along with sensitivity and specificity for predicting outcomes. These metrics were chosen as they are commonly accepted measures for correlation (R^2) and accuracy/validity (sensitivity and specificity) while also being the most reported metrics across studies.

We assessed individual study risk of bias (ROB) and applicability to the research question separately and in duplicate using the Quality Assessment of Diagnostic Accuracy Studies (QUADAS) II tool (6). We assessed ROB for each included study based on patient selection, use of index test, reference standard and flow and timing. We assessed applicability for each included study based on patient selection, use of index test and reference standard. Using this tool, each domain was rated as either "low," "unclear," or "high" ROB. Whereas "high" ratings in any two or more of seven domains led to an overall rating of "high" ROB. A high rating in one domain, or an "unclear" rating in more than one of the seven domains led to an overall rating of "intermediate" ROB (6). "Low" ratings in all domains, or "unclear" in a single domain, were judged to be at "low" ROB.

Data Analysis

Given the expected clinical and methodological heterogeneity, we did not plan to pool data generated from this systematic review and rather protocolized qualitative and narrative summary of the data.

RESULTS

We included 45 observational studies that met eligibility criteria (2, 7–50) (**Fig. 1**). **Supplementary Table 2** (https://links.lww.com/CCM/H873) presents a summary of the included studies. Of these, 15 studies included children, and 30 studies evaluated adults. Most studies enrolled patients from a medical critical care unit, but there was one study that enrolled purely cardiac surgical unit patients (20), another surgical ICU patients (19), another pediatric necrotizing enterocolitis patients (29) and one that included pediatric burn unit patients (23). Two of the included studies imputed PF ratios from SF ratios using various statistical techniques and compared this to the actual PF ratios (36, 37), whereas the remainder compared measured SF ratios to measured PF ratios.

Study Quality and Risk of Bias assessments

Thirty-two of 45 studies were judged to be at low ROB, with 12 studies being judged at intermediate ROB and one study being judged at high ROB (**Supplementary Table 3**, https://links.lww.com/CCM/H873). Some studies did not clearly report how individual patients were identified for study inclusion and were downgraded to unclear ROB for patient selection. We also downgraded studies (mostly abstracts) for flow and timing that did not report exactly at what time Spo_2 and Pao_2 values were taken, in relation to each other. Both the index test and reference standards for studies were objective, providing numerical findings that prevented any biased interpretations. Finally, we downgraded one study (29) for applicability reasons as it only reported on patients with burns.

Relationship Between SF and PF Ratio

Although no sensitivity analyses was performed, among studies that stratified Spo_2 values, SF to PF imputation was less accurate when the Spo_2 was equal to or greater than 97%. Most studies were able to establish a linear relationship between SF and PF ratio with R^2 statistics of correlation ranging from 0.5 to 0.8. A few studies established a logarithmic relationship between SF and PF (9, 40, 47, 50) and the remainder established a variety of nonlinear relationships including

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.

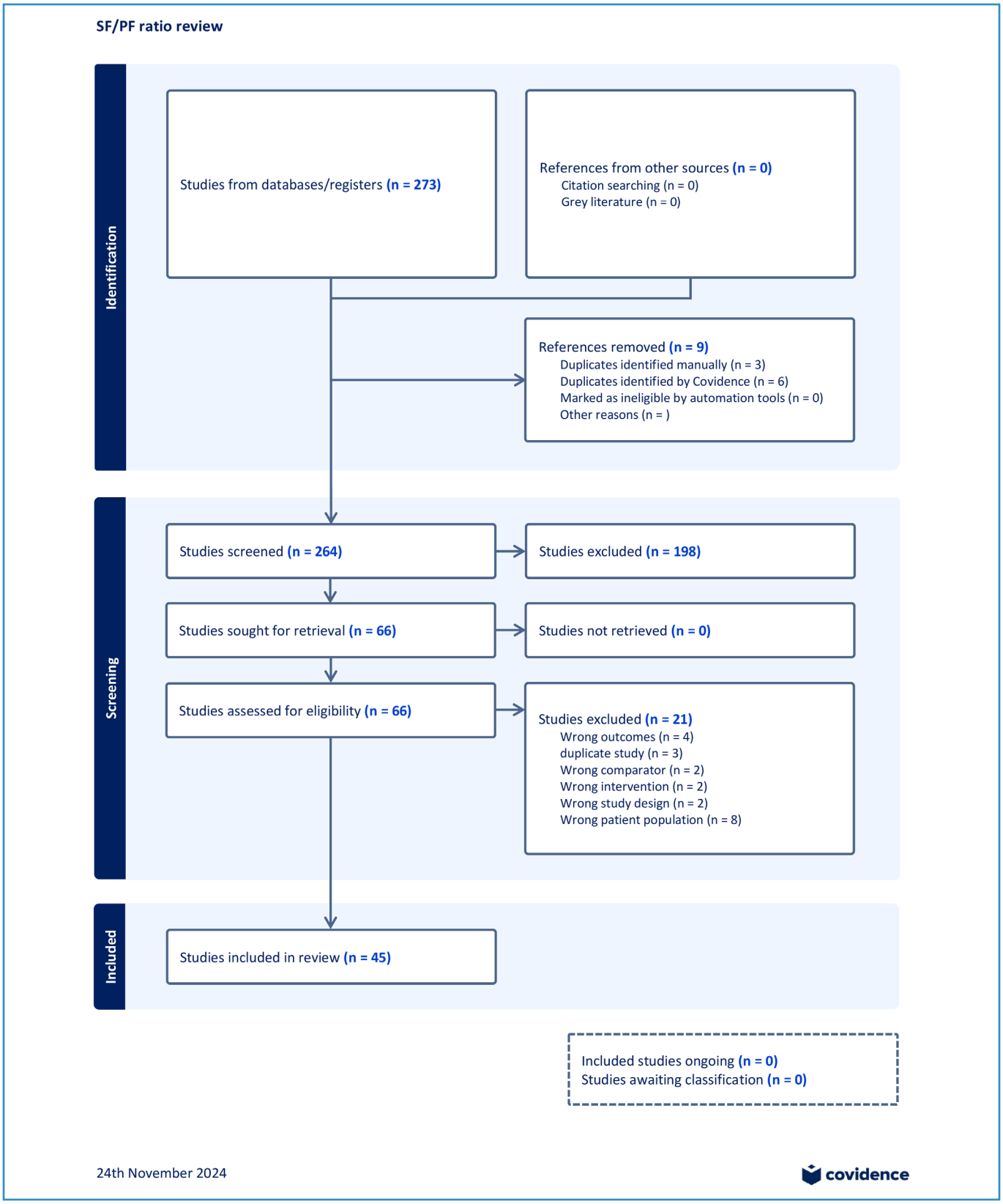

Figure 1. Preferred Reporting Items for Systematic reviews and Meta-Analyses flow diagram.

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.

TABLE 1.
Spo_2/Fio_2 Pao_2/Fio_2 Equations

Author	Equation	SF		
		PF = 300	PF = 200	PF = 100
Rice et al (2) (linear)	$SF = 64 + 0.84 \times PF$	315	235	148
Schmidt et al (40) (logarithmic)	$Log(PF) = -0.15 + 1.07 \times Log(SF)$	295	195	102
Beauchamp et al (44) (nonlinear)	$PO_2 = \left(\frac{27.8^{2.8}}{\frac{1}{S}-0.99}\right)^{\frac{1}{2.8}}$	Dependent on Fio_2 and Spo_2		
Ellis (nonlinear)	$PO_2 = \{11700/(1/S-1)+[50^3+(11,700/1/S-1))^2]^{1/2}\}^{1/3}+\{11,700/(1/S-1))-[50^3+(11,700/(1/S-1))^2]^{1/2}\}^{1/3}$	Dependent on Fio_2 and Spo_2		

PF = Pao_2/Fio_2, Po_2 = partial pressure of arterial oxygen, S = peripheral oxygen saturation, SF = Spo_2/Fio_2.
Displays the full equation to convert SF to PF, along with the corresponding SF for the key PF ratios of 100, 200, and 300.

inverse functions (15), the Ellis equation (8, 10, 37), logistical regressions (29, 48), and custom algorithms using neural nets (36, 44) and other methods (35). Of the included studies, 21 reported discrimination capacity of the SF ratio to differentiate between a single ARDS severity state (e.g., PF < 200 vs. PF > 200) and 7 reported discrimination capacity between multiple ARDS severity states (2, 11, 15, 19, 21, 22, 39, 40).

Although all studies established a correlational relationship between SF and PF ratios, there was no clear best equation for imputing PF ratios from SF ratios. In the hopes of highlighting the studies with the most clinically informative results, we applied the following criteria: size (prioritizing larger studies), methodology (prioritizing those that were most rigorously performed and most accurate), frequency of use (prioritizing equations that were used across multiple different studies), ease of use (prioritizing equations that were easy to apply at the bedside), and generalizability (prioritizing those that enrolled a wide range of ICU patients across centers). To assess studies for methodology and accuracy, we focused on those that were published in full, those that had a high degree of correlation, sensitivity and specificity and those that made adjustment for positive end-expiratory pressure (PEEP) and Spo_2 accuracy. We developed these criteria iteratively and after we had extracted data.

Based on these criteria, we have prioritized four specific SF to PF imputation methods, which we describe below. In addition, **Table 1** illustrates corresponding SF values for the commonly used PF values of 300, 200, and 100 using these equations.

Linear Imputation. *Equation Number 1.* Enrolled patients from the ARDS Network tidal volume trial (51) as the derivation data set and the ARDS Network PEEP trial (52) as the validation data set. Using multiple, large, well designed, multicenter randomized control trials for data collection, authors developed a simple, easy to use (at the bedside) and accurate linear equation, that was used in multiple studies. Of note, some studies (37) suggest that nonlinear imputation methods may be more accurate.

Rice et al (2), adult patients with ARDS, *n* equals to 2673 measurements

$$SF = 64 + 0.84 \times PF$$

where R^2 equals to 0.89; SF 315 discriminates for PF 300 with a sensitivity of 91%, specificity of 56% and AUC of 0.88; SF 235 discriminates for PF 200 with a sensitivity of 85%, specificity of 85% and AUC of 0.93.

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.

Log Linear Imputation. *Equation Number 2.* Used a well-validated database (Multiparameter Intelligent Monitoring in Intensive Care II) that represented data collected from patients in a tertiary care hospital (Beth Israel Deaconess Medical center) and over a 7-year period. This study had the second largest number of measurements and presented an accessible imputation method. It also was performed across a diverse population of critically ill ventilated patients (medical/surgical, trauma, transplant and cardiac). Limitations include the complexities of applying a logarithmic equation at the bedside, the fact that only a single center is represented questioning generalizability and relatively lower accuracy compared with other methods.

Schmidt et al (40), mechanically ventilated patients, *n* equals to 7544 measurements

$$\mathrm{Log}\,(\mathrm{PF}) = -0.15 + 1.07 * \mathrm{Log}\,(\mathrm{SF})$$

where R^2 equals to 0.72; SF 295 discriminates for PF 300 with a sensitivity of 99%, specificity of 10% and AUC of 0.73; Kappa agreement score for ARDS diagnosis using PF vs. SF cutoff was 0.87.

Nonlinear Equations. *Equation Number 3.* This equation was derived using two independent databases from two tertiary care centers including noncardiac PICU patients (53). This imputation method was found to be highly accurate using multiple neural network models, derived from a large number of measurements and investigators were more diligent to ensure that $\mathrm{Pa}_{\mathrm{O}_2}$, $\mathrm{FI}_{\mathrm{O}_2}$, and $\mathrm{Sp}_{\mathrm{O}_2}$ measurements were performed at the same point in time. This equation is limited by complexity, the involvement of only two centers and the inclusion of only pediatric patients, which may limit generalizability.

Beauchamp et al (44), Sauthier et al (53) PICU patients, *n* equals to 17,239 measurements

$$\mathrm{PO}_2 = \left(\frac{27.8^{2.8}}{\frac{1}{S} - 0.99}\right)^{\frac{1}{2.8}}$$

$\mathrm{P}_{\mathrm{O}_2}$ equals to partial pressure of arterial oxygen, S equals to peripheral oxygen saturation. Kappa agreement score for hypoxemia classification was 0.71, mean absolute error was 61, AUC for PF less than 100 was 0.9; demonstrated to perform well/better against other nonlinear equations including Ellis equation (53).

Equation Number 4 (Ellis equation). A variety of studies used this equation which is the inverse derivation of the Severinghaus equation that describes a relationship between $\mathrm{Pa}_{\mathrm{O}_2}$ and $\mathrm{Sp}_{\mathrm{O}_2}$ (add ref for the Severinghaus equation). This equation was most often studied in comparison to other linear and nonlinear equations. This equation has been shown to be highly accurate compared with others (37). However, the main limitation is its complexity which limits its usage at the bedside or incorporation into prognostic tools.

Brown et al (42), Philippe et al (17), Ortiz et al (15) mechanically ventilated patients, *n* equals to 1034 measurements

$$\mathrm{PO}_2 = \left\{\frac{11,700}{\frac{1}{S} - 1} + \left[50^3 + \left(\frac{11,700}{\frac{1}{S} - 1}\right)^2\right]^{1/2}\right\}^{1/3} + \left\{\frac{11,700}{\frac{1}{S} - 1} - \left[50^3 + \left(\frac{11,700}{\frac{1}{S} - 1}\right)^2\right]^{1/2}\right\}^{1/3}$$

$\mathrm{P}_{\mathrm{O}_2}$ equals to partial pressure of arterial oxygen, S equals to peripheral oxygen saturation. R^2 = 0.85 (Philippe et al [9]); mean absolute error was 30.2 for $\mathrm{Sp}_{\mathrm{O}_2}$ less than 96%, AUC for PF less than 150 was 0.95, AUC for PF less than 300 was 0.87 (Brown et al [37]).

The Utility of SF and PF Ratios at Predicting Clinical Outcomes

Data comparing the effectiveness of SF to PF ratios at predicting clinical outcomes in critically ill patients with respiratory failure was limited. One large cohort study (9) performed 4728 measurements in patients with ARDS or those undergoing general anesthesia and generated a logarithmic equation to convert SF ratios to PF ratios. Then they examined the validity of using these converted SF ratios to predict SOFA scores, ICU length of stay and ventilator-free days. They found that SOFA scores using the SF ratio correlated strongly with SOFA scores using the traditional PF ratio (R^2 = 0.85). In addition, they found that using the SF ratio (as compared with PF ratio) to inform the SOFA score led to similar associations with ICU length of stay (R^2 0.26 vs. 0.36, p = 0.005) and ventilator-free days (R^2 –0.28 vs. –0.33, p = 0.025). Another observational study (24) compared the prognostic ability of SF, derived by the Rice equation (Equation Number 1 above) vs. PF ratio in 362 patients with ARDS (2) and found no difference

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.

in clinical outcomes including hospital mortality, duration of mechanical ventilation and duration of ICU stay. One study (27) of 2796 patients presenting to the emergency department with sepsis found that using the SF-informed SOFA score predicted ICU mortality with sensitivity of 91% and specificity of 51% which was very similar to the PF-informed SOFA score (sensitivity 90%, specificity 50%). When considering the pediatric population (32), a study of mechanically ventilated children used a previously derived linear equation (31) to compare outcomes in patients who had ARDS diagnosed by SF criteria vs. PF criteria. Similar to when using the PF ratio, this study demonstrated a clear stepwise increase in mortality as SF ratio decreased. Interestingly, a number of the included studies (32, 49, 50) showed that SF ratio may be better at predicting mortality than PF in critically ill patients with respiratory failure, whereas one (41) did not.

DISCUSSION

We have summarized the existing evidence evaluating the relationship between SF and PF ratios and explored how well SF ratio can be used in place of the PF ratio in the clinical setting. We have demonstrated that the SF ratio has a moderate to strong correlation with the PF ratio, and may have similar prognostic value, particularly in the context of ARDS.

Despite this strong correlation, substituting the SF ratio in place of the PF ratio must be done with caution. Prior work by Erlebach et al (54) shows that while the SF ratio performs well at categorizing moderate to severe ARDS, it is less reliable at properly categorizing no or mild ARDS. Further, changes in FIO_2 disproportionately affect the SF ratio when compared with changes in SpO_2 (54), resulting in the SF ratio being more reflective of treatment decisions rather than the disease state. Furthermore, multiple variables play a role in determining how SpO_2 and PaO_2 are related, including but not limited to PEEP, skin tone, skin and end organ perfusion, hemoglobin, metabolic state (pH, CO_2 levels, temperature), body mass index, and age (2, 9). Finally, the range of possible values for PaO_2 are much larger than SpO_2, as evidenced by the flat parts of the oxyhemoglobin disassociation curve. As a result, the SpO_2 to PaO_2 conversion is inaccurate when SpO_2 is above 97%.

Given the above, it is currently not possible to develop a conversion method or equation that can enable perfect interchangeability between the SF and PF ratios. As a result, there are multiple possible equations with moderate to strong correlational coefficients ($R^2 > 0.5$) (55, 56) and good discrimination of ARDS severity states which would be appropriate for converting SF ratios to PF ratios. However, from a practical standpoint, linear equations are least complex and easiest to apply at the bedside. Although some nonlinear equations, may have comparable (or even slightly better) accuracy to linear equations, their complexity may preclude more widespread clinical use. Thus, a linear equation, like the one provided by Rice et al (2) with a high R^2 value (representing a well fitted model) may be the most optimal equation for bedside use.

Although establishing a correlational relationship between PF and SF ratios is important, ultimately, clinicians likely care most about their ability to diagnose, prognosticate and impact management decisions for patients. Multiple studies demonstrate that the SF ratio may be just as good, if not better, than the PF ratio at predicting mortality and other patient important outcomes (9, 11, 24, 27, 32, 50). Given the ease of measurement, especially in resource-limited settings, the SpO_2 is more frequently used to manage patients than PaO_2. However, limitations such as poor perfusion states, dark skin tone, and low hemoglobin necessitate the use of PaO_2 in certain clinical contexts (4). Although the PF ratio is currently the more accepted measure for facilitating the diagnosis of ARDS and for inclusion in SOFA scores (3, 57), the SF ratio may be a justifiable and more convenient alternative.

The performance of SF ratios across a variety of clinical contexts should be further investigated. For one, the accuracy of SF ratios in situations where the SpO_2 may not accurately measure tissue perfusion (low flow states, dark skin color, etc.) remains relatively uncertain. This could be both due to inaccuracies in the SpO_2 measure itself, but also due to clinician induced changes in the FIO_2 to respond to the erroneous SpO_2. The relationship of perfusion, SF ratios and PF ratios in patients undergoing cardiovascular surgeries and in hypothermic states are also less clear. Finally, in mechanically ventilated patients, varying levels of PEEP can alter SpO_2 and PaO_2 measurements independent of FIO_2. Although one study (9) showed a limited role

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.

of PEEP in the relationship between SF and PF ratios, further research addressing this question is needed.

To our knowledge, this is the first systematic review that comprehensively examines the relationship between SF and PF ratios in critically ill patients. Strengths of this review include the comprehensive search across multiple databases, pre-registration of the protocol and the inclusion of 45 studies and tens of thousands of PF/SF ratio comparisons. This review also has limitations. Although we point out a few priority equations based on relatively objective criteria, the lack of one primary equation prevents standardization of measurements across clinical settings. In particular, PF ratios are used most often as cutoff values for the diagnosis of ARDS or for the prognosis of critically ill patients through inclusion in the SOFA score. Without one primary equation, it becomes challenging to generate a set of corresponding SF values for set PF thresholds. Given the heterogeneity in studies, we were unable to comprehensively compare the performance of SF ratios as compared with PF ratios for prognostication of patient important outcomes such as mortality, length of stay, etc. Furthermore, the data were limited to patients where clinical judgement had led to arterial sampling.

INTERPRETATION

SF and PF ratios demonstrate moderate to strong correlation, and may have similar prognostic value. However, using SF to PF ratios interchangeably must be done with caution, particularly in patients with Spo_2 greater than 97% and/or non-severe ARDS. Although there is no clear optimal method to convert SF to PF ratios, linear equations show acceptable correlation and are most easily applied at the bedside.

This study was conceived by working members of the Sequential Organ Failure Assessment 2.0 group, including Drs. Chaudhuri, Pekkarinen, Sendagire, Martin, Jung, Lazarte, and Rochwerg. Screening and data collection was performed by Drs. Chaudhuri, Lazarte, Shah, and Pitre. The article was written by Drs. Chaudhuri and Rochwerg. All authors edited and reviewed the article.

ACKNOWLEDGMENTS

We thank Karin Dearness, MLIS, and Kaitryn Campbell, MLIS, for generation and review of the search strategy.

1 *Department of Medicine, McMaster University, Hamilton, ON, Canada.*

2 *Department of Critical Care, St. Joseph's Healthcare Hamilton, Hamilton, ON, Canada.*

3 *McMaster University, Hamilton, ON, Canada.*

4 *Division of Respirology, Department of Medicine, University of Toronto, Toronto, ON, Canada.*

5 *Division of Intensive Care, Department of Anaesthesiology and Intensive Care, University of Helsinki and Helsinki University Hospital, Helsinki, Finland.*

6 *Makerere University College of Health Sciences, Kampala, Uganda.*

7 *D'Or Institute for Research and Education, Rio de Janeiro, Brazil.*

8 *Division of Pulmonary, Allergy, Critical Care and Sleep Medicine, Emory University School of Medicine, Atlanta, GA.*

9 *Grady Memorial Hospital, Atlanta, GA.*

10 *Department of Cardiology, Pulmonology and Vascular Medicine, Heinrech-Heine-University Dusseldorf, Dusselsorf, Germany.*

11 *Cardiovascular Research Institute Dusseldorf, Heinrech-Heine-University Dusseldorf, Dusselsorf, Germany.*

12 *Anaesthesia and Intensive Care Medicine, School of Medicine, University of Galway, Galway, Ireland.*

13 *Department of Anaesthesia and Intensive Care Medicine, Galway University Hospital, Galway, Ireland.*

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's website (http://journals.lww.com/ccmjournal).

Dr. Martin received funding from Grifols, UpToDate, Elsevier, and Eagle Pharmaceuticals. The remaining authors have disclosed that they do not have any potential conflicts of interest.

For information regarding this article, E-mail: dipayan.chaudhuri@medportal.ca

REFERENCES

1. Orraine L, Are BW, Atthay AM: The acute respiratory distress syndrome. *N Engl J Med* 2000; 342:1334–1349
2. Rice TW, Wheeler AP, Bernard GR, et al; National Institutes of Health, National Heart, Lung, and Blood Institute ARDS Network: Comparison of the Spo_2/Fio_2 ratio and the Pao_2/Fio_2 ratio in patients with acute lung injury or ARDS. *Chest* 2007; 132:410–417
3. Ranieri VM, Rubenfeld GD, Thompson BT, et al; ARDS Definition Task Force: Acute respiratory distress syndrome: The Berlin definition. *JAMA* 2012; 307:2526–2533
4. Martin D, Johns C, Sorrell L, et al: Effect of skin tone on the accuracy of the estimation of arterial oxygen saturation by pulse oximetry: A systematic review. *Br J Anaesth* 2024; 132:945–956
5. Blanchet MA, Mercier G, Delobel A, et al: Accuracy of multiple pulse oximeters in stable critically ill patients. *Respir Care* 2023; 68:565–574

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.

6. Whiting PF, Rutjes AWS, Westwood ME, et al; QUADAS-2 Group: QUADAS-2: A revised tool for the quality assessment of diagnostic accuracy studies. *Ann Intern Med* 2011; 155:529–536
7. Wongsrichanalai V: Correlation between the Spo_2/Fio_2 ratio and Pao2/Fio2 ratio in ARDS. *Chest* 2019; 155:98A
8. Philippe M, Nadia A, Davide BS, et al: Pao_2/Fio_2 ratio estimation from Spo_2/Fio_2 ratio: A planned companion study of the SPECTRUM study. French Intensive Care Society, International Congress Reanimation 2018. Paris, France, January 2018, p. 13
9. Pandharipande PP, Shintani AK, Hagerman HE, et al: Derivation and validation of Spo_2/Fio_2 ratio to impute for Pao_2/Fio_2 ratio in the respiratory component of the Sequential Organ Failure Assessment score. *Crit Care Med* 2009; 37:1317–1321
10. Ortiz G, Bastidas A, Garay-Fernández M, et al: Correlation and validity of imputed Pao_2/Fio_2 and Spo_2/Fio_2 in patients with invasive mechanical ventilation at 2600 m above sea level. *Med Intensiva* 2022; 46:501–507
11. Serpa Neto A, Cardoso SO, Ong DSY, et al: The use of the pulse oximetric saturation/fraction of inspired oxygen ratio for risk stratification of patients with severe sepsis and septic shock. *J Crit Care* 2013; 28:681–686
12. Muniraman HK, Kibe R, Namjoshi A, et al: Evaluation of correlation and agreement between Spo_2/Fio_2 ratio and Pao_2/Fio_2 ratio in neonates. *J Pediatr Intensive Care* 2022; 4:267–358
13. Lohano PD, Baloch SH, Gowa MA, et al: Correlation between the ratio of oxygen saturation to fraction of inspired oxygen and the ratio of partial pressure of oxygen to fraction of inspired oxygen in detection and risk stratification of pediatric acute respiratory distress syndrome. *Cureus* 2021; 13:e18353
14. Lobete Prieto C, Medina Villanueva A, Modesto i Alapont V, et al: Predicción del índice Pao_2/Fio_2 a partir del índice Spo_2/Fio_2 ajustado por la medición transcutánea de CO_2 en niños críticamente enfermos. *Anales de Pediatría* 2011; 74:91–96
15. Lobete C, Medina A, Rey C, et al: Correlation of oxygen saturation as measured by pulse oximetry/fraction of inspired oxygen ratio with Pao_2/fraction of inspired oxygen ratio in a heterogeneous sample of critically ill children. *J Crit Care* 2013; 28:538.e1–538.e7
16. Ahmed Z, Naz F, Raza AB, et al: Correlation of Pao_2/Fio_2 ratio with Spo_2/Fio_2 ratio in children on mechanical ventilation. *Pak J Med Health Sci.* 2021; 15:3069–3071
17. Aslanidis T, Myrou A, Chytas E, et al: Relation between Pao_2/Fio_2 ratio, Spo_2/Fio_2 ratio, oxygenation index and ventilation ratio in critically ill patients. *Crit Care* 2013; 17:P92
18. Solomon R. Azamfirei L. Copotoiu R, et al: Evaluation of the Pao_2/Fio_2 ratio versus Spo/Fio ratio in patients with acute respiratory distress syndrome. *Rom J Anaesth Intensive Care* 2010; 17:87–91
19. Babu S, Abhilash KPP, Kandasamy S, et al: Association between Spo_2/Fio_2 ratio and Pao_2/Fio_2 ratio in different modes of oxygen supplementation. *Indian J Crit Care Med* 2021; 25:1001–1005
20. Bashar FR, Vahedian-Azimi A, Farzanegan B, et al: Comparison of non-invasive to invasive oxygenation ratios for diagnosing acute respiratory distress syndrome following coronary artery bypass graft surgery: A prospective derivation-validation cohort study. *J Cardiothorac Surg* 2018; 13:1–8
21. Bhiri M. Tl. JNZ. BIK. GI. EF. KM. MT: Contribution of Spo_2/Fio_2 and Pao_2/Fio_2 ratios in patients with COVID-19 acute respiratory distress syndrome. *Intensive Care Med Exp* 2022; 10:39
22. Bilan N, Dastranji A, Behbahani AG: Comparison of the Spo_2/Fio_2 ratio and the Pao_2/Fio_2 ratio in patients with acute lung injury or acute respiratory distress syndrome. *J Cardiovasc Thorac Res* 2015; 7:28–31
23. Cambiaso-Daniel J, Voigt CD, Rivas E, et al: Correlation between Pao_2/Fio_2 and peripheral capillary oxygenation/Fio_2 in burned children with smoke inhalation injury. *Pediatr Crit Care Med* 2017; 18:e472–e476
24. Chen W, Janz DR, Shaver CM, et al: Clinical characteristics and outcomes are similar in ARDS diagnosed by oxygen saturation/Fio_2 ratio compared with Pao_2/Fio_2 ratio. *Chest* 2015; 148:1477–1483
25. Cinesi-Gómez C, García-García P, López-Pelayo I, et al: Correlación entre la saturación de oxihemoglobina por pulsioximetría y la presión arterial de oxígeno en pacientes con insuficiencia respiratoria aguda. *Revista Clínica Española* 2017; 217:522–525
26. Fukuda Y, Tanaka A, Homma T, et al: Utility of Spo_2/Fio_2 ratio for acute hypoxemic respiratory failure with bilateral opacities in the ICU. *PLoS One* 2021; 16:e0245927
27. Guirgis F, Sterling S, Smotherman C, et al: 1456: A comparison of sofa using Spo_2/Fio_2 ratio to sofa, SIRS, and QSOFA for sepsis mortality. *Crit Care Med* 2016; 44:439–439
28. James EJG, t S, Palaniappan P, et al: A prospective study to compare the ratio of Spo_2/Fio_2 with Pao_2/Fio_2 ratio among critically ill children requiring respiratory support. *Pediatr Crit Care Med* 2018; 19:232–232
29. Sim NZ, Hall NJ, Peters MJ, et al: Comparison of Spo_2/Fio_2 and Pao_2/Fio_2 based markers of respiratory dysfunction in neonates with necrotising enterocolitis. *Intensive Care Med* 2013; 39:1–200
30. Jusi RL, Limpin E, Bayot R, et al: Determination of critical threshold value of Spo_2/Fio_2 ratio in the diagnosis of acute lung injury. *Eur Respir J* 2012; 40(Suppl 56):P2020
31. Khemani RG, Patel NR, Bart RD, et al: Comparison of the pulse oximetric saturation/fraction of inspired oxygen ratio and the Pao_2/fraction of inspired oxygen ratio in children. *Chest* 2009; 135:662–668
32. Khemani RG, Rubin S, Belani S, et al: Pulse oximetry vs. Pao_2 metrics in mechanically ventilated children: Berlin definition of ARDS and mortality risk. *Intensive Care Med* 2015; 41:94–102
33. Kibe RN. Muniraman H. Fletcher K, et al: Correlation of Pao_2/Fio_2 ratio to Spo_2/Fio_2 ratio in neonates with respiratory distress. *J Investig Med* 2018; 66:113–114
34. Kumar A, Aggarwal R, Khanna P, et al: Correlation of the Spo_2/Fio_2 (S/F) ratio and the Pao_2/Fio_2 (P/F) ratio in patients with COVID-19 pneumonia. *Med Intensiva* 2022; 46:408–410
35. Lee B, Singh G, Bartock J, et al: Hiding in plain sight: Sao_2/Fio_2 as the superior hypoxic measure. *Intensive Care Med Exp* 2020; 8:73
36. Tabary M, Ren S, Zupetic J, et al: Validation of machine learning algorithms in predicting Pao_2 from Spo_2 in mechanically ventilated patients. *Am Thorac Soc Int Conf Meet Abstr Am Thorac Soc Int Conf Meet Abstr* 2022; 205:A5401–A5401
37. Brown SM, Grissom CK, Moss M, et al; NIH/NHLBI PETAL Network Collaborators: Nonlinear imputation of Pao_2/Fio_2

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.

from Spo_2/Fio_2 among patients with acute respiratory distress syndrome. *Chest* 2016; 150:307–313

38. Xia Y, Xu Q, Guo Z, et al: A modified screening protocol for ARDS in patients with respiratory support based on Spo_2 and Fio_2: A single-center prospective, observational study. *J Intensive Med (PG-)* 2024; 5:51–57
39. Singla S, Bhargava S, Pooni PA, et al: Comparison of Spo_2-with Pao_2-based indices of oxygenation in children with respiratory failure. *J Pediatr Intensive Care (PG-7)* 2024; 7:1–107
40. Schmidt MFS, Gernand J, Kakarala R: The use of the pulse oximetric saturation to fraction of inspired oxygen ratio in an automated acute respiratory distress syndrome screening tool. *J Crit Care* 2015; 30:486–490
41. Shah N, Sarangi BU, Shankar GH, et al: Comparison of Pao_2/Fio_2 (PF ratio) to Spo_2/Fio_2 (SF ratio) and OI to OSI for predicting short term outcomes in children with acute hypoxemic respiratory distress: A prospective observational study. *Indian J Pediatr (PG-)* 2024; 92:823–829
42. Jain R, Jain A, Jain S, et al: Correlation of oxygenation indices in invasive mechanical ventilated adult patients. *Lung India*. 2024; 41:176–180
43. Gandhi V, Trivedi L, Mulani K, et al: Comparison of Spo_2/Fio_2 (SF) ratio and Pao_2/Fio_2 (PF) ratio in mechanically ventilated patients with ARDS in COVID ICU for prognosis of ARDS: A prospective, observational study. *Res J Med Sci*. 2023; 17:12–16
44. Beauchamp FO, Sauthier M: A continuous and noninvasive method to estimate Pao_2 and PF ratio: A brief report. *Crit Care Med* 2024; 52:S142
45. Alur TR, Iyer SS, Shah JN, et al: A prospective observational study comparing oxygen saturation/fraction of inspired oxygen ratio with partial pressure of oxygen in arterial blood/fraction of inspired oxygen ratio among critically ill patients requiring different modes of oxygen supplement. *Indian J Crit Care Med* 2024; 28:251–255
46. Sdiri I, Doghri HH, Klai I, et al: Spo_2/Fio_2 ratio as an assessment tool for hypoxemia evaluation in intensive care department. *Ann Intensive Care* 2023; 13:50
47. Schmidt M, Gernand J, Kakarala R. Correlation between SF ratio and PF ratio in an unselected mechanically ventilated ICU population | A25. Lung injury: Markers and outcomes. American Thoracic Society International Conference, San Diego, May 2014
48. Sathiyakumar A, Pitrowsky M, Davidson M, et al: 499: Developing a predictive model for assessing Pao_2/Fio_2 using Sao_2/Fio_2 in critically ill patients. *Crit Care Med* 2011; 39 (Suppl):A499
49. Santana A, Sousa J de, Amorim F, et al: Sao_2/Fio_2 ratio as risk stratification for patients with sepsis. *Crit Care* 2013; 17:P51
50. Ray S, Rogers L, Pagel C, et al: Pao_2/Fio_2 ratio derived from the Spo_2/Fio_2 ratio to improve mortality prediction using the pediatric index of mortality-3 score in transported intensive care admissions*. *Pediatr Crit Care Med* 2017; 18:e131–e136
51. Yndrome S, Etwork N: Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. *N Engl J Med* 2000; 342:1301–1308
52. Brower RG, Hopkins Uni-versity J, Lanken PN, et al: Higher versus lower positive end-expiratory pressures in patients with the acute respiratory distress syndrome. *N Engl J Med* 2004; 351:327–336
53. Sauthier M, Tuli G, Jouvet PA, et al: Estimated Pao_2: A continuous and noninvasive method to estimate Pao_2 and oxygenation index. *Crit Care Explor* 2021; 3:e0546
54. Erlebach R, Pale U, Beck T, et al: Limitations of Spo_2/Fio_2-ratio for classification and monitoring of acute respiratory distress syndrome—an observational cohort study. *Crit Care* 2025; 29:1–9
55. Hinkle D, Wiersma W, Jurs S. *Applied Statistics for the Behavioural Sciences*. 5th ed. Boston, Houghton Mifflin, 2003
56. Schober P, Schwarte LA: Correlation coefficients: Appropriate use and interpretation. *Anesth Analg* 2018; 126:1763–1768
57. Singer M, Deutschman CS, Seymour C, et al: The Third International Consensus Definitions for Sepsis and Septic Shock (Sepsis-3). *JAMA*. 2016; 315:801–810

Copyright © 2026 by the Society of Critical Care Medicine and Wolters Kluwer Health, Inc. All Rights Reserved.